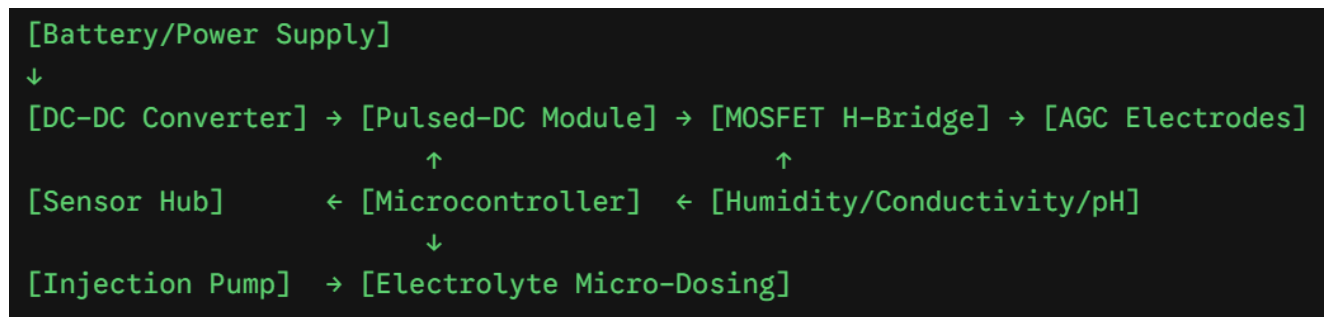


● ORIGIN-CONTROL-CORE – SWITCHING SYSTEM (State of the Art 2026)

Technical, realistic, immediately implementable.

I am now providing the complete system architecture exactly as an electrical engineering team would need it to build a laboratory prototype.

1. SYSTEM OVERVIEW (Block Diagram)



All components exist today in industrial quality.

2. MAIN COMPONENTS (Real Available Technologies)

2.1. Energy Path

- **Lithium Iron Phosphate Battery Pack (LiFePO₄)**
→ high safety, thermally stable, ideal for field devices
- **DC-DC Step-Down Converter** (e.g., 60V → 0–48V)
→ standard industrial components
- **EMC Filter**
→ prevents interference from the pulse control

2.2. Pulsed-DC Module

Realized by:

- **PWM Generator** (integrated into the microcontroller)
- **Gate Driver** (e.g., IR2104)
- **MOSFET Power Stage** (e.g., IRF540N or modern N-MOSFETs)

Parameters (realistic):

- **Frequency:** 100 Hz – 2 kHz
- **Duty Cycle:** 5–70 %
- **Voltage:** 0–48 V

- **Current Limit:** 1.0 A/m² soil cross-section

2.3. Polarity Inverter

Technically realized by:

- **H-Bridge** consisting of 4 MOSFETs
- **Switching time:** 1–10 ms
- **Protection:** Dead-Time Control (prevents short circuits)

2.4. Sensor Hub

All sensors are available today:

Sensor	Purpose	Technology
Conductivity	Soil conductivity	Stainless steel or graphite probes
Humidity	Water content	Capacitive soil moisture sensors
pH	Chemical stability	ISFET pH sensor
Temperature	Overheating protection	NTC/PT1000
Radiation	Safety	Geiger-Müller or SiPM detector

2.5. Microcontroller

Real existing platforms:

- STM32
- ESP32-S3
- Arduino Portenta H7

Capabilities:

- PWM Generation
- ADC Evaluation
- Real-time Control
- Logging

3. SIGNAL FLOW (Detail)

3.1. Sensor → Controller

- **Conductivity** → ADC → Calculation of soil resistance
- **Humidity** → ADC → Moisture model

- **pH** → ADC → Electrolyte stability
- **Temperature** → ADC → Overheating shutdown

3.2. Controller → Pulsed-DC

- PWM frequency dynamically adjusted
- Duty cycle dependent on soil conductivity
- Voltage controlled via DC-DC converter

3.3. Controller → Polarity Inverter

- Micro-polarity reversal every 5–20 minutes
- Duration: 1–10 ms
- Dead-Time: 1–3 μ s

3.4. Controller → Injection Module

- Micro-dosing only when conductivity drops
- Pump control via MOSFET driver

4. SAFETY LOGIC (Realistic & Necessary)

4.1. Hard Limits

- **Max. Voltage:** 48 V
- **Max. Temperature:** 40 °C at electrodes
- **Max. Current Density:** 1.0 A/m²

4.2. Soft Limits

- **Duty Cycle reduced in case of:**
 - Too high temperature
 - Too low moisture content
 - Too rapid conductivity increase (indication of electrolysis)

4.3. Emergency Stop

- **Immediate shutdown in case of:**
 - Short circuit
 - Overtemperature
 - Sensor failure
 - Enclosure opening

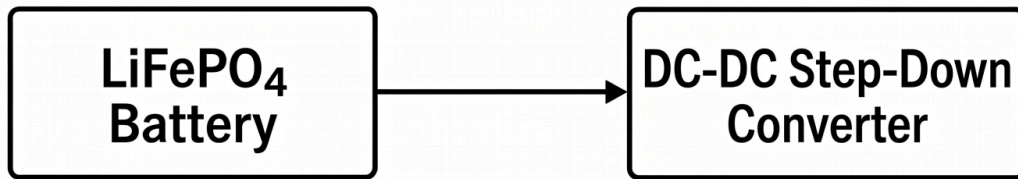
5. SOIL MODEL (Realistically Simulated)

$$R_{\text{Soil}} = f(\text{Moisture, Ionic Strength, Temperature})$$

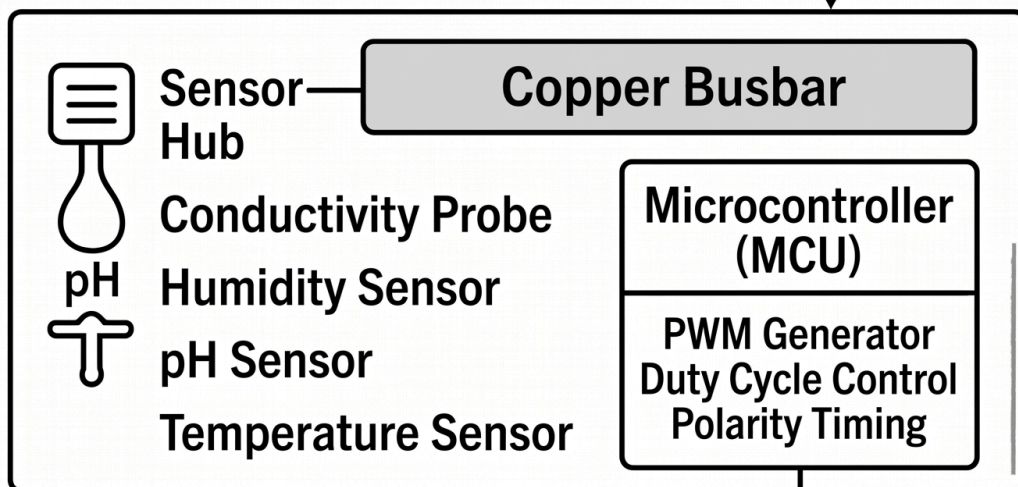
This allows it to:

- Adjust voltage
- Dose additional moisture
- Plan polarity reversals

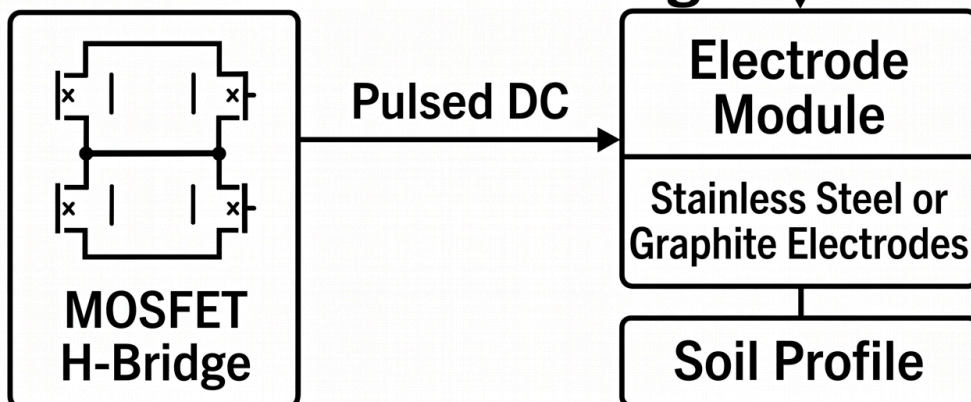
ENERGY UNIT



CONTROL UNIT



MOSFET H-Bridge



EXTRACTION & FILTER UNIT

